InventOR: A Prompt-Configured, No-Fine-Tuning LLM Workflow for Material-Level Inventory Control

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Abstract

This paper presents *InventOR*, a prompt-configured large-language-model (LLM) workflow for material-level inventory decision support under demand and lead-time uncertainty. A material-scoped historical CSV and inline parameter block are supplied through an enterprise LLM application; no project-specific model weights were trained or fine-tuned on enterprise data. The LLM returns a policy artifact that deterministic parsing and simulation evaluate against an SAP-derived comparator and an SAP-safety-lead-time-informed operations-research (r, Q) comparator. The LLM is therefore a candidate-policy generator, not an autonomous ERP controller. The executed artifact set covers all 365 eligible plant-material pairs in a three-plant industrial dataset after retry handling and capacity validation. A common-feasible cohort of 346 pairs applies identical MOQ and per-material storage rules to all arms. The post-cutoff simulation reports 77.77% demand-weighted aggregate fill, 98.11% mean material fill, and 3,002.40 mean simulated material-horizon holding/order cost for the LLM-emitted arm. Working-day-arrival sensitivity preserves the service–

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cost ordering. These values remain descriptive simulated comparison rather than controlled production-trial evidence. Stored metadata do not establish the exact deployed prompt or model/runtime identity; the contribution is a transparent prototype workflow and disclosure of controls required for prospective evaluation.
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1. Introduction

Supply chain planning operates under volatile demand, uncertain replenishment, and disruption exposure [1]. Many material-control decisions nevertheless still depend on simple parameters: reorder points, safety stock, order quantities, review periods, and sometimes safety lead time; these settings strongly affect service and inventory exposure [2, 3]. Classical operations research offers rigorous tools for stochastic and robust inventory decisions [4, 5, 6], but operational planners often need a narrower capability: converting imperfect historical records into transparent, defensible, and auditable material-level policy values [7].

Large language models (LLMs) can help connect business intent, operational evidence, and analytical workflows [8, 9]. In inventory control, however, recommendations require explainability, reproducibility, accountability, and meaningful human control before planners can act on them [10, 11]. This creates a practical design requirement for hybrid AI-supported OR systems: LLM artifacts must be inspectable, while their numeric recommendations require explicit simulation and human release controls [9, 11].

This paper addresses that requirement through *InventOR*, an operations-research-first workflow for material-level inventory decision support. An enterprise LLM application produces policy artifacts from material-scoped inputs; deterministic software parses and simulates their numeric values. The study evaluates independent plant-material pairs, not shared-capacity inventory control, and does not estimate implementation cost or planner adoption.

The paper makes three contributions to applied AI/OR methodology and inventory-control practice.

C1 Prompt-configured architecture. Material-specific inputs pass through a configured enterprise LLM application without project-specific training or fine-tuning, with deterministic scoreability checks and planner review [12].

C2 Executable boundary. The paper documents the material-input contract, alias-aware conversion, capacity validation, and simulation boundary used in the evaluation.

C3 Complete exploratory evidence. All 365 eligible pairs from 233,078 parsed records yielded scoreable, capacity-feasible artifacts after retry handling; 346 pairs meet the common feasibility rules used for the three-arm descriptive comparison.

The remainder of the paper reviews related work, presents the executed workflow and policy logic, reports exploratory post-cutoff simulation results, and discusses the controls needed before a valid deployment evaluation.

2. Literature Review

Operations research has long addressed supply-chain uncertainty through stochastic programming, robust optimization, and scenario-based planning [4, 5, 6]. These methods formalize variability and recourse, but operational inventory teams often face a more direct implementation problem: maintaining reorder points, safety stock, safety lead time, and order quantities from incomplete historical evidence in ERP-driven planning environments [13, 14, 2, 3]. Static rules can be too coarse, while sophisticated models can be difficult to explain and maintain. This motivates decision support that keeps OR discipline while making material-level parameter setting more transparent and auditable [7, 15].

Recent LLM research proposes natural-language interfaces, tool-using agents, and pipelines that translate natural-language requests into optimization model structures or executable code [9, 16, 17, 18]. Published conceptual systems and recent preprints illustrate a rapidly evolving frontier, including multi-agent LLM systems that make autonomous per-period inventory decisions [19, 20, 21]. They do not by themselves specify how inventory histories should be reduced into demand regimes, lead-time diagnostics, service buffers, and feasible policy parameters. Prior work therefore supports the hybrid position adopted here: LLM artifacts can structure workflows and explanations, while deterministic parsing and simulation make their consequences inspectable and planners retain decision authority [9, 10].

InventOR follows this hybrid view. It supplies one filtered historical CSV for the focal material plus an inline parameter block through an enterprise LLM application. The intended prompt specification prohibits stochastic simulation, machine-learning libraries, and unrestricted data fabrication, but the stored run metadata do not independently identify the deployed prompt or runtime. Because LLM outputs can contain fabricated or unfaithful content [22], the executed downstream parser uses alias-aware lookups and nested-field resolution before requiring numeric positive reorder point and order quantity, numeric non-negative safety stock, and a resolvable simulator family. No project-specific model fitting or fine-tuning is performed. The design treats the LLM as an artifact-generation component inside a bounded decision process. Table 1 positions the framework relative to recent AI-supported OR and inventory-control work.

Table 1: Literature positioning: InventOR relative to recent AI-supported OR contributions for inventory and supply chain decision support.

Study	**LLM role**	**Policy class**	**Demand model**	**Analytical result**	**Evaluation**
Huang et al. [16]	NL→code	General OR	Not specified	None	Benchmark
Ahmaditeshnizi et al. [18]	NL→model	General OR	Not specified	Solver-linked	Benchmark
Zhang et al. [20]	Reasoning agent	General OR	Not specified	Model and solve	Case studies
Quan and Liu [21]	Multi-agent control	Inventory	Sequential demand	Per-period decisions	Simulation
Prak et al. [23]	None	Inventory control	Compound Poisson / intermittent	Robust estimation	Empirical study
Babai et al. [24]	None	Forecasting-linked inventory	Regime-specific	Review synthesis	Review
Wasserkrug et al. [9]	Interface	General	Not specified	None	Conceptual
Cohen et al. [17]	Vision	General SCM	Not specified	None	Vision statement
InventOR	**Prototype**	**Fixed-quantity simulator**	**Artifact-defined values**	**Descriptive**	**Post-cutoff simulation**

The table highlights the paper's narrower contribution: material-level parameter artifacts evaluated by a fixed-quantity simulator. Unlike InvAgent's sequential multi-agent decisions, InventOR emits one stationary policy per material from a bounded historical package; unlike OptiMUS and OR-LLM-Agent, it does not formulate or solve general optimization models.

3. Methodology

The unit of analysis is a plant-material pair. The framework records a chain of custody from historical records to planner-facing policy values by separating deterministic input preparation, configured LLM inference, schema parsing, deterministic simulation, and human review [7, 25]. Each LLM session receives one filtered, material-scoped historical CSV as a session resource and a small inline parameter block containing current cost, MOQ, service, SAP reference, and material-level storage-capacity fields. No project-specific model weights are trained, fine-tuned, or updated. Because the stored metadata omit prompt hash, application version, and model/runtime identifier, this study does not claim that the exact deployed instruction configuration is independently reproducible.

3.1. Problem Definition: Single-Item Continuous-Review Control with Stress-Adjusted Service Buffering

The executed task is a single-item, single-echelon continuous-review simulation with lost sales [13, 14, 26]. Each plant-material pair is treated independently; multi-item coupling, shared capacity, substitution, and production scheduling are outside the pilot. For the SAP-SLT-informed OR comparator, pre-cutoff working-day demand yields mean demand $\bar{d}_i$, annualized demand $\bar{D}_i^{\mathrm{ann}} = 250\bar{d}_i$, lead-time-demand dispersion, and stationary (R_i, Q_i) parameters. This comparator is not independent of ERP inputs because source SAP safety lead time enters effective lead time. For the

SAP-derived arm, source safety stock is retained while R_i and Q_i are recomputed by the same executable routines. For the LLM arm, scoreable R_i, Q_i, and SS_i^{policy} values originate directly in the artifact and are converted by the permissive parser; the reference equations below do not reconstruct them. The executed simulator uses only the resulting reorder point and order quantity. Policy labels and narrative fields are audit context rather than separate simulator implementations.

Reference policy calculation. The following equations describe the executed SAP-SLT-informed OR comparator; they do not reconstruct direct LLM-emitted policies. The order quantity $Q_i > 0$ is rounded upward to an integer, floored at source MOQ, and then projected to its material-level storage limit when possible. The reorder point $R_i \geq 0$ triggers replenishment. The EOQ anchor minimizes the standard annual cost approximation

$$\min_{Q_i} TC_i\Big(Q_i; SS_i^{\text{policy}}\Big) = h_i\left(\frac{Q_i}{2} + SS_i^{\text{policy}}\right) + K_i\frac{\bar{D}_i^{\text{ann}}}{Q_i},$$

where $h_i = \texttt{holding\_cost\_rate}_i \cdot \texttt{unit\_cost}_i$ and $K_i = \texttt{ordering\_cost}_i$. The implementation first uses $Q_i = \max\{\text{MOQ}_i, \lceil\sqrt{2K_i(250\bar{d}_i)/h_i}\rceil\}$ when demand and ordering cost are positive, and MOQ otherwise. For every arm, a supplied capacity requires $\text{MOQ}_i \leq Q_i$ and $SS_i + Q_i \leq \texttt{max\_storage\_units}_i$; comparator quantities above the residual limit are reduced, while pairs with $SS_i + \text{MOQ}_i$ above the limit are excluded from the common cohort. The SAP-SLT-informed OR safety stock is $SS_i = \lceil z_{\alpha_i}\sigma_{LTD,i}\rceil$, with $R_i = \lceil \bar{d}_i L_i^{\text{eff}} + SS_i\rceil$. The SAP-derived arm substitutes source SAP safety stock into the same reorder-point expression. No lot-size multiple, stress coefficient, or shortage penalty is executed.

Core constraints. Inventory evolves under a lost-sales continuous-review rule on the working-day decision set $\mathcal{W}$. All arms share the first observed post-cutoff on-hand inventory; the SAP-SLT-informed OR $R_i + Q_i$ is used only when that observation is unavailable. The initial on-order pipeline is empty. Duplicate same-material, same-date rows are aggregated by summing demand, receipts, corrections, and scrap and averaging positive lead times, but the simulator consumes only demand and excludes observed receipts, corrections, and scrap. The intended prompt describes source lead time as working days, whereas the executable simulator rounds its mean and schedules every modeled order by that number of calendar days:

$$I_{i,t} = \max(0,\ I_{i,t-1} + a_{i,t} - d_{i,t}), \qquad o_{i,t} = Q_i \cdot \delta_{i,t}, \quad t \in \mathcal{W},$$

where $a_{i,t}$ is the quantity whose rounded calendar-day due date is no later than row date t. Scheduled arrivals are added first, realized demand is served subject to lost sales, and the post-demand inventory position is computed as on-hand plus outstanding orders. A fixed quantity is ordered when

$$\delta_{i,t} = \mathbf{1}\{\mathrm{IP}^{+}_{i,t} \le R_i\}, \qquad \mathrm{IP}^{+}_{i,t} = I_{i,t} + \sum_{s:\,\mathrm{due}(s)>t} o_{i,s},$$

with $o_{i,t} = Q_i \delta_{i,t}$ scheduled after rounded mean lead time. The parser may resolve an artifact label to an order-up-to family, but the current simulator still places fixed Q_i and does not execute a distinct order-up-to rule. Parameters remain stationary. Shared observed initialization and calendar-day receipt offsets are part of the executed comparison design.

The reorder point targets a nominal cycle service level $\alpha_i \in [0.5, 0.999]$

after numeric clamping. Under the normal approximation,

$$\mathbb{P}\Big(D_{i,L_i^{\text{eff}}} \le R_i\Big) \ge \alpha_i \iff R_i \ge \bar{d}_i L_i^{\text{eff}} + z_{\alpha_i}\, \sigma_{LTD,i}.$$

The reported backtest metric is the ex-post fill rate, which is distinct from the nominal cycle-service target because it also depends on Q_i and realized demand [27].

Lead-time-demand variance is propagated under the demand and lead-time independence approximation,

$$\sigma_{LTD,i}^2 = L_i^{\text{eff}} \sigma_{D,i}^2 + \bar{d}_i^2 \sigma_{L,i}^2.$$

The executed SAP-SLT-informed OR buffer is

$$SS_i^{\text{OR}} = z_{\alpha_i} \sigma_{LTD,i},$$

with values rounded upward:

$$SS_i^{\text{OR}} = \lceil z_{\alpha_i} \sigma_{LTD,i} \rceil, \qquad R_i = \left\lceil \bar{d}_i L_i^{\text{eff}} + SS_i^{\text{OR}} \right\rceil.$$

The executable statistics add non-negative source SAP safety-lead-time days to mean lead time when constructing comparator lead-time-demand dispersion (Section 3.5 discusses this confound). LLM-emitted values remain those emitted and converted by the permissive parser.

Parser and review boundary. Artifact policy labels are audit context rather than executable routing instructions. The parser resolves scoreable numeric values and a simulator family, while the evaluated simulator applies a fixed-quantity replenishment rule. It does not execute label-specific optimization or deterministic escalation. Planner review remains required before any operational use.

Methodologically, the contribution is therefore not to replace OR with an LLM. Rather, it is an integration of deterministic input preparation, configured LLM artifact generation, schema parsing, and explicit policy simulation in a single prototype workflow [10, 11].

3.2. Historical Input Schema and Canonical Analysis Table

The source table is grouped by plant and material, but the uploaded material CSV contains only the subset listed in Table 2. Plant and material identifiers are carried by the run key and inline block rather than repeated in the uploaded rows. The inline block also passes current cost, MOQ, service, SAP reference, working-day demand, lead-time, and material-level storage-capacity variables. Deterministic preprocessing supplies input normalization, eligibility filters, and simulator statistics. Stock-flow fields omitted from the uploaded subset are not analyzed by the executed LLM path.

Table 2: Fields included in the material-scoped CSV uploaded by the executed runner.

Column	Type	Description and method use
`date`	date / datetime	Calendar observation date; time ordering, aggregation, and diagnostics
`actual_demand`	numeric	Realized demand; mean, variability, intermittency, trend, and regime classification
`delivery_time`	numeric	Recorded replenishment lead time; described as working days in the intended prompt but interpreted as calendar-day offsets by the simulator
`minimum_order_quantity`	numeric	Minimum replenishment quantity; order-size feasibility constraint
`safety_stock_units`	numeric	Existing safety stock; benchmarking against computed recommendations
`safety_time_workdays_actual`	numeric	Source SAP safety-lead-time field; currently enters comparator effective lead time
`supplier_plant_distance`	numeric	Supplier-to-plant distance; contextual lead-time-risk information
`reliability_calc`	numeric	Source supplier or lane reliability context
`transportation_network`	string	Source transportation-network context
`service_level_target`	numeric	Source service target used by comparator calculations
`unit_cost`	numeric	Purchase or standard cost per unit in the source dataset's internal cost denomination; inventory-value proxy and input to holding-cost derivation and EOQ calculation
`holding_cost_rate`	numeric	Annual carrying-cost rate as a decimal (e.g., 0.20 for 20% of unit value); used to derive the annual cost of carrying one unit in stock, balance replenishment frequency against stock exposure, and compute economic order quantity
`ordering_cost`	numeric	Fixed replenishment cost in the source dataset's internal cost denomination; input to EOQ calculation and simulated material-horizon holding/order cost
`max_storage_units`	numeric	Material-level upper bound applied to all evaluated arms: MOQ $\leq Q$ and $SS + Q \leq$ `max_storage_units`
`is_working_day`	binary	Working-day flag; working-day demand statistics and non-zero demand share

3.3. Analytical Workflow

The workflow transforms a historical material snapshot into a policy artifact through deterministic preparation, LLM inference, parsing, and simulation (Figure 1).

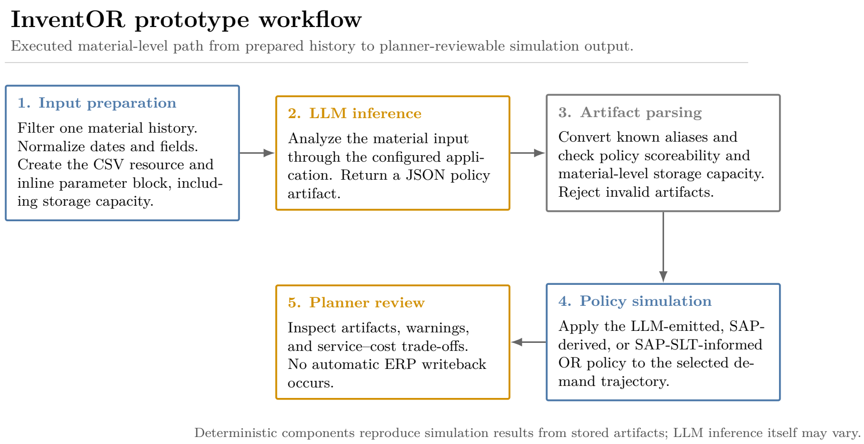

Figure 1: Prototype workflow from material input preparation to policy artifact and simulation output.

The runner uploads a normalized material CSV and inline block; the configured application returns a JSON artifact. The parser converts accepted fields to the canonical safety-stock, reorder-point, and order-quantity interface and rejects malformed or infeasible outputs. Missing or invalid artifacts are recorded and retried. The simulator evaluates accepted LLM values, SAP-derived values, and the SAP-SLT-informed OR policy on the common horizon. An aggregate-only manifest records active-input, evaluation-code, and stored-artifact hashes; it explicitly records unavailable deployed prompt, application, model/runtime, setting, and timestamp metadata. Stored inputs, artifacts, and code reproduce a recorded simulation, but not a fresh LLM completion; planners retain implementation authority.

3.4. Artifact Completeness and Future Readiness

The completeness audit asks whether each eligible pair has a scoreable artifact after retries; it does not estimate first-pass reliability, explanation completeness, or deployment readiness. The post-cutoff simulation uses artifacts generated exclusively from pre-cutoff inputs and is descriptive rather than a controlled prospective trial [28, 29]. A future evaluation must freeze the deployed configuration, model/runtime identifiers, access controls, and approval workflow.

3.5. Safety Lead Time Computation

Safety lead time (SLT) is an ancillary ERP-facing design field. The prototype specification proposes combining transport, supplier, and quality lead-time variability with a reliability-deficit premium [13, 3], but the present release does not document estimators for those components. In the executed comparator code, non-negative source SAP safety-lead-time days are added to mean lead time before computing both SAP-derived and OR-computed dispersion and reorder points (this coupling is also flagged in Section 3.1). The OR arm is therefore labelled SAP-SLT-informed, and no separate SLT effect is identified.

3.6. Post-Cutoff Policy Simulation

The confidential operational extract contains 233,078 parsed CSV records, 411 plant-material pairs, and three anonymized plants (Plant A, Plant B, and Plant C). The parameter cutoff is 2026-03-10 and the post-cutoff simulation horizon ends on 2026-08-27. Eligibility filters retain 365 plant-material pairs with sufficient pre-cutoff and post-cutoff evidence: 234 from Plant A, 65 from Plant B, and 66 from Plant C. Plant C has no positive source working-day flags, so its calendar flags are derived from same-date Plant A and Plant B votes; ties classify as working days. This deterministic repair permits a common decision calendar but lacks an alternative-calendar sensitivity analysis. SAP-derived and SAP-SLT-informed OR parameters use pre-cutoff information. The LLM artifacts reported here were regenerated from cutoff-bounded pre-cutoff inputs, so every arm is evaluated on the same post-cutoff horizon. Nineteen pairs cannot meet comparator safety stock plus MOQ within their supplied capacity; these are excluded before the common 346-pair comparison.

The simulated arms are: (i) SAP-derived, retaining source SAP safety stock while recomputing reorder point and EOQ order quantity from pre-cutoff statistics; (ii) SAP-SLT-informed OR, using the same statistics, source safety-lead-time input, and feasibility rules but without LLM inference; and (iii) LLM-emitted, the artifact converted into the canonical safety-stock, reorder-point, and order-quantity interface. The simulator is continuous-review and lost-sales: all arms begin with the same first observed post-cutoff on-hand inventory, using the OR $ROP + Q$ only as a shared fallback. The initial on-order pipeline is empty. On each working row, modeled arrivals due after the rounded mean calendar-day lead time are received, demand is served subject to lost sales, post-demand inventory position is checked, and an order is placed if the threshold is met. Observed post-cutoff receipts, corrections, and scrap are excluded. This common startup design may create cutoff transients. Metrics are fill rate, aggregate demand-weighted fill rate, materials above 95% fill, zero-stockout materials, stockout days, average on-hand inventory, and simulated material-horizon holding/order cost.

For policy p and material i, the cost summaries use the simulated horizon of T_i working days. Let $\overline{I}_{i,p}$ be average on-hand inventory, h_i annual holding cost per unit, K_i fixed ordering cost, and $n_{i,p}^{\text{orders}}$ simulated orders. The implementation uses a constant 250 annual working days and computes

$$TC_{i,p} = h_i \overline{I}_{i,p} \frac{T_i}{250} + K_i n_{i,p}^{\text{orders}}.$$

This is a simulated material-horizon holding/order cost in the source dataset's internal cost denomination, not an audited annual financial or booked accounting result. Table 3 reports arithmetic means across the cohort. No explicit shortage penalty is included, so cost values must be read jointly with fill rate and stockout days.

4. Results

4.1. Completion and Artifact Validity

The final execution produced parser-scoreable artifacts for all 365 eligible plant-material pairs. The simulation artifact reports zero missing and zero parser-rejected outputs after retry handling. The parser uses compatibility containers, alias lookups, and nested-field resolution before requiring numeric positive reorder point and order quantity, numeric non-negative safety stock, a resolvable simulator family, and compliance with the supplied material-level storage bound. This establishes policy scoreability and capacity feasibility for the deterministic simulator, not full conformance to every field requested by the prompt. Eventual completeness after targeted retries is not an estimate of first-pass reliability or production availability.

4.2. Main Three-Arm Exploratory Simulation

Table 3 reports three policy arms on the same 346 common-feasible material trajectories. The SAP-derived arm retains source SAP safety stock but recomputes reorder point and EOQ quantity from pre-cutoff statistics; it is not the unchanged incumbent SAP policy. SAP-SLT-informed OR evaluates an (r, Q) comparator whose effective lead time retains source SAP safety lead time. The LLM-emitted arm evaluates artifact values generated from cutoff-bounded pre-cutoff inputs after permissive conversion and material-level capacity validation. All arms satisfy the same MOQ and supplied per-material storage rule. Nineteen otherwise eligible pairs whose comparator safety stock plus MOQ exceed capacity are excluded. The comparison remains descriptive simulation rather than controlled production trial.

Table 3: Post-cutoff simulation on 346 common-feasible plant-material pairs. All arms use pre-cutoff parameterization or cutoff-bounded inputs and satisfy identical MOQ and supplied storage rules. Cost is mean simulated material-horizon holding/order cost; no shortage penalty is included. SO days is the total count of stockout days across the cohort.

Policy	n	**Mean FR (%)**	**Agg. FR (%)**	**≥95% (%)**	**Zero SO (%)**	**Mean cost**	**SO days**
SAP-derived	346	98.55	80.25	95.95	84.10	3,346.87	699
SAP-SLT OR	346	98.39	79.84	95.38	91.91	4,016.81	754
LLM-emitted	346	98.11	77.77	93.93	86.71	3,002.40	860

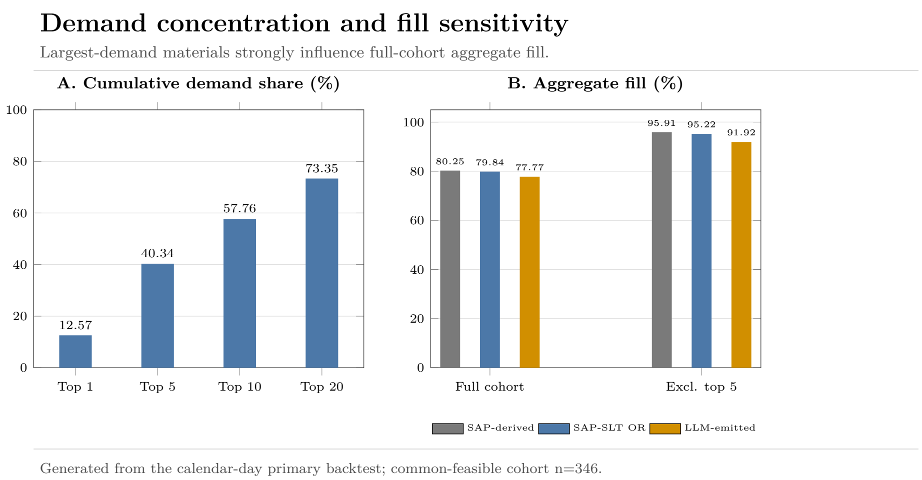

Figure 2: Demand concentration and aggregate-fill sensitivity. Panel A reports cumulative demand share for the largest materials. Panel B compares full-cohort aggregate fill with the value after excluding the five largest-demand materials.

Relative to SAP-derived, LLM-emitted policies have 2.48 percentage points lower aggregate fill, 344.47 lower mean cost, and 161 additional stockout days. Relative to SAP-SLT-informed OR, they have 2.07 percentage points lower aggregate fill, 1,014.40 lower mean cost, and 106 additional stockout days. These values describe one simulated cohort, not a definitive ranking.

Aggregate evidence is demand-concentrated. The largest 1, 5, 10, and 20 materials contribute 12.57%, 40.34%, 57.76%, and 73.35% of post-cutoff demand. Excluding the five largest-demand materials raises aggregate fill to 91.92% for LLM-emitted, 95.91% for SAP-derived, and 95.22% for SAP-SLT-informed OR, versus 77.77–80.25% in the full cohort (Figure 2). Thus, the largest-demand items drive aggregate fill and should not be obscured by cohort averages.

At material level, LLM-emitted fill is not lower than SAP-derived for 321 of 346 materials (92.8%), while its cost is no higher for 235 (67.9%); both conditions hold for 214 (61.8%). Against SAP-SLT-informed OR, the corresponding counts are 318 (91.9%), 304 (87.9%), and 279 (80.6%). Paired LLM-minus-comparator fill differences have first quartile, median, and third quartile of 0.00 percentage points for both comparators, reflecting many exact fill ties. Corresponding cost differences are -267.54, -44.00, and 38.23 versus SAP-derived and -816.03, -203.67, and -24.30 versus SAP-SLT-informed OR. These distribution summaries and aggregate results jointly show service–cost trade-offs rather than uniform policy ranking.

The artifact audit covers all 365 scoreable policies: normalized policy-family agreement is 90.68%, and 42 artifacts meet a prespecified diagnostic-outlier rule (CV absolute error above 0.75 or LLM-to-pipeline safety-stock ratio outside [0.5, 2.0]). Removing the 23 flagged artifacts present in the common cohort raises LLM aggregate fill from 77.77% to 79.80%, while SAP-derived reaches 80.33% and SAP-SLT-informed OR reaches 79.94%; this post-hoc diagnostic subset does not establish a causal correction. Replacing calendar-day receipt offsets with working-day offsets gives aggregate fills of

77.53%, 79.99%, and 79.58%, respectively. This sensitivity preserves the observed trade-off, but does not resolve empty-pipeline, excluded-flow, or calendar-repair limitations.

5. Discussion

The current results document an applied AI/OR prototype on one anonymized industrial dataset. The LLM artifacts are generated from cutoff-bounded, pre-cutoff material histories, and all arms use the same post-cutoff horizon. The SAP-SLT-informed OR comparator is not independent of ERP inputs. The comparison remains descriptive rather than a controlled prospective trial: the 365-material artifacts support auditing and code-path verification, while the 346-pair common-feasible cohort supports no definitive performance ranking.

InventOR makes the input, artifact, and evaluation boundary inspectable without making LLM inference deterministic [10, 9]. An aggregate-only manifest hashes the active input, evaluation code, and stored artifacts, while explicitly identifying unavailable deployed prompt, application, model/runtime, setting, and timestamp records. A stored input, artifact, and simulator version reproduce a recorded result; a fresh completion may vary. The tracked historical 315-material snapshots are byte-identical rather than distinct generation replicates, so they cannot estimate generation variance. Missing-output lists, parser checks, retries, and material-level simulation rows keep incomplete execution visible [11, 30]. The workflow avoids project-specific training, but offers no evidence on setup effort, inference cost, or planner adoption.

The accepted policies satisfy a per-material storage bound, not a joint capacity constraint. They do not model shared capacity, substitution, or coordinated ordering. Their service–cost trade-off is not a replacement recommendation; a prospective trial with paired uncertainty intervals and enforced planner review is required before adoption [13, 23].

Several limitations delimit the conclusions. All comparisons remain descriptive simulations without confidence intervals or formal significance tests. Evidence comes from one anonymized dataset, one horizon, and one preserved full-cohort LLM generation; fresh completions may vary. An earlier unconstrained 315-material artifact set is retained for provenance, but its different cohort and validation end date preclude attributing differences to the storage rule. Demand is concentrated, with the five largest materials representing 40.34% of demand. The working-day-arrival and diagnostic-outlier sensitivities are limited checks, not replacements for a pipeline or flow-data reconstruction. The simulation begins with observed on-hand inventory but no outstanding-order pipeline, excludes observed receipts, corrections, and scrap, and uses calendar-day receipt offsets in primary results. Plant C's working-day calendar is reconstructed without an alternative-calendar sensitivity. The SAP-SLT-informed OR arm retains an SAP input. The capacity rule is per material and does not model shared capacity, substitution, multi-echelon interactions, or coordinated ordering [31, 32]. Recommendation quality remains dependent on data quality, historical representativeness, application configuration, and policy conversion rules [15, 25]. Narrative completeness and planner effectiveness were not evaluated.

These limitations define the scope for future work: a prospective, paired-uncertainty evaluation that freezes the deployed configuration and model/runtime identifiers, initializes the replenishment pipeline, validates lead-time and calendar semantics, removes SAP inputs from an independent

OR comparator, reports first-pass reliability, and extends to multi-item and shared-capacity settings with planner review. The present evidence supports a prototype with traceable inputs, stored artifacts, and deterministic simulation; it does not support a controlled-trial comparative performance claim [28, 29].

6. Conclusion

This paper presents *InventOR*, an applied AI/OR prototype that treats LLMs as candidate-policy generators within deterministic preparation, parsing, and simulation. The framework preserves an inspectable link between material histories, policy fields, and simulated outcomes without project-specific model training or fine-tuning.

The executed artifact set contains 365 scoreable policies generated from cutoff-bounded, pre-cutoff material histories. Every accepted LLM policy satisfies its material-level storage bound. On the 346-pair cohort where all arms satisfy identical MOQ and supplied storage rules, LLM-emitted aggregate fill is 77.77%, mean material-level fill is 98.11%, mean simulated material-horizon cost is 3,002.40, and stockout days total 860. Working-day receipt scheduling gives the same service–cost ordering. These values are descriptive simulated-comparison evidence, not confidence-interval estimates or controlled prospective-trial evidence. The framework establishes a reviewable prototype and protocol for subsequent prospective evaluation, not a deployment or adoption claim.

Author Contributions

Conceptualization, Methodology, Software, Formal analysis, Validation, Visualization, Writing – original draft, Writing – review and editing: Aris Dressino.

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Declaration of Competing Interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and Code Availability

The raw operational data and material-level artifacts cannot be publicly released because they contain commercially sensitive material, supplier, and planning information. A confidentiality-screened release package containing analysis code, the anonymized variable schema, aggregate evaluation results, manuscript sources, and aggregate integrity manifest is available at `github.com/DressPD/inventor_public`. No archival DOI has been assigned. The deployed application configuration, model/runtime identifier, inference dates, and prompt hashes were not preserved and are not claimed as available.

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Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

During manuscript preparation, the author used OpenAI GPT-5.6 through OpenCode for language editing, consistency checks, citation review, and LaTeX proofing support. The author reviewed and edited the content after use and takes full responsibility for the content of the published article. Figures were rendered deterministically from author-reviewed LaTeX and Python source; generative AI did not directly create or manipulate image content. The research workflow evaluated in the paper is itself the object of study and is described separately in the methodology.

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